

CS 3430 Database Introduction

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What Is Data?

Is This Data?

1365.81
1247.18
318.457
252.1648
202.91

Yes,
but it is meaningless.

With Attribute

Population
1365.81
1247.18
318.457
252.1648
202.91

We know better,
but still not very useful!

With Another Variable

Place	Population (millions)
China	1365.81
India	1247.18
USA	318.457
Indonesia	252.1648
Brazil	202.91

Much better,
but need more!

Meaningful Information

Place	Population (millions)	Date (as of)
China	1365.81	July 27, 2014
India	1247.18	July 27, 2014
USA	318.457	July 27, 2014
Indonesia	252.1648	July 1, 2014
Brazil	202.91	July 27, 2014

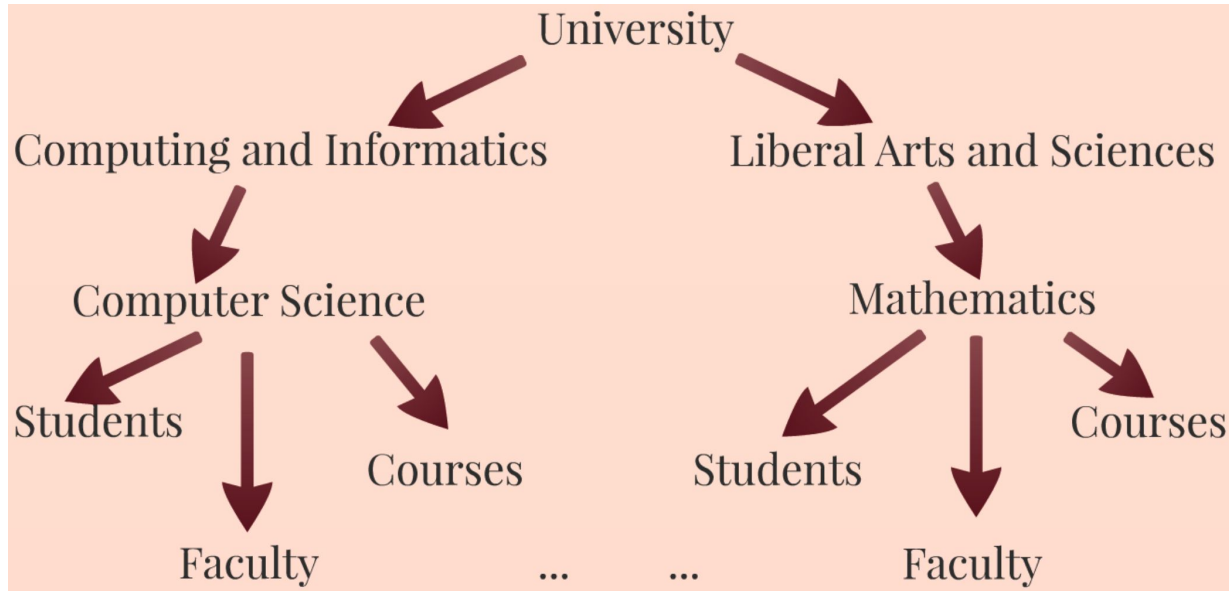
What Is Data?

- Known facts that can be recorded and have an implicit meaning
- Data is information that has been translated into a form that is efficient for movement or processing.
- Numbers, words, measurements, descriptions ...
- Need a way to **store** and **retrieve** data in correct, convenient, efficient, and secure manner

** It is acceptable for the word **data** to be used as both singular and plural.*

Traditional File Processing System

- Data is organized into files and folders



An Example Of A File Processing System



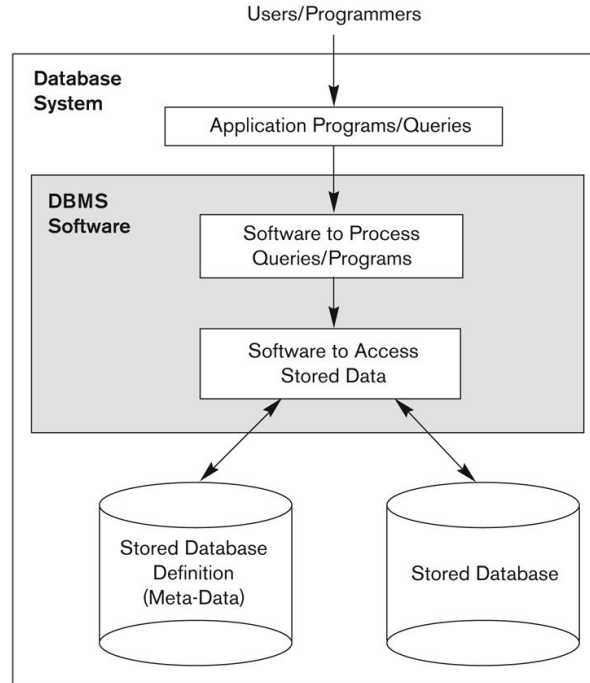
Issues

- Data redundancy and inconsistency
- Difficulty in retrieving specific data
- Data integrity concerns
- Security problem

Database – Basic Definitions

- Database
 - A collection (structure) of related data
- Database Management System (DBMS)
 - A software package/system to facilitate the creation and maintenance of a computerized database
- Database System
 - The DBMS software, together with the data itself and sometimes applications

Simplified Database System Environment



Database Management System (DBMS)

- A general-purpose software that controls all access to database
- Create, update, and delete data
- Ensure correctness, security, convenience, and efficiency
- Manage concurrent access by multiple users
- Handle and recover from errors

DBMS Used In Our Class



Application

- A program that accesses the database by sending queries to DBMS
- Queries:
 - that access different parts of data and formulate the result of a request
- Transactions:
 - that may read some data and update certain values or generate new data
- Database system:
 - The DBMS software together with the data itself. Sometimes, the applications are also included.
- Applications must NOT allow unauthorized users to access data

Miniworld

- Or **Universe of Discourse (UoD)**
- Some aspects of the real world, which is represented by a database.
- Changes to the miniworld are reflected in the database.

An Example (The Conceptual Data Model)

- Mini-world for the example:
 - Part of a University environment
- Some mini-world entities:
 - STUDENTs
 - COURSEs
 - SECTIONs
 - DEPARTMENTS
 - INSTRUCTORs

An Example (The Conceptual Data Model)

- Some mini-world relationships:
 - SECTIONS are of specific COURSEs
 - STUDENTs take SECTIONS
 - COURSEs have prerequisite COURSEs
 - INSTRUCTORs teach SECTIONS
 - COURSEs are offered by DEPARTMENTs
 - STUDENTs majors in DEPARTMENTs

An Example

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Insulation Between Programs And Data

- Program-data independence
 - The structure of data files is stored in the DBMS catalog separately from the access programs.
- Program-operation independence
 - User application programs can operate on the data by invoking operations through their names and arguments, regardless of how the operations are implemented.

Multiple Views Of The Data

- Each user may see a different view of the database, which describes only the data of interest to that user.

TRANSCRIPT

Student_name	Student_transcript				
	Course_number	Grade	Semester	Year	Section_id
Smith	CS1310	C	Fall	08	119
	MATH2410	B	Fall	08	112
Brown	MATH2410	A	Fall	07	85
	CS1310	A	Fall	07	92
	CS3320	B	Spring	08	102
	CS3380	A	Fall	08	135

COURSE_PREREQUISITES

Course_name	Course_number	Prerequisites
Database	CS3380	CS3320
		MATH2410
Data Structures	CS3320	CS1310

Data Sharing And Transaction

- Concurrent access
- Concurrency control
- OLTP (Online Transaction Processing)
- ACID
 - Atomicity
 - Consistency
 - Isolation
 - Durability

Advantages Of Using DBMS

- Controlling redundancy
- Restricting unauthorized access
- Providing persistent storage for program objects
- Providing storage structures for efficient query processing
- Providing backup and recovery
- Providing multiple user interfaces
- Representing complex relationships among data
- Enforcing integrity constraints

When Not To Use A DBMS

- Overhead costs of using a DMBS
 - High initial investment in hardware, software, and training
 - Overhead for providing security, concurrency control, recovery, and integrity functions
- When a DMBS may be unnecessary
 - If the database and applications are simple, well defined, and not expected to change
 - If access to data by multiple users is not required
- When a DBMS may be infeasible
 - In embedded systems where a general-purpose DBMS may not fit in available storage

Database System

Data Abstraction

- A data model hides storage details and presents users with a conceptual view of the database.
- Programs refer to the data model constructs rather than data storage details.

Data Models

- A set of concepts to describe the **structure** of a database, the **operations** for manipulating these structures, and certain **constraints** that the database should obey.

Data Models

- Structure (Constructs)
 - Typically include elements (and their data types), groups of elements (entity, record, table...), and relationship
- Constraints
 - Specify some restrictions on valid data
 - Must be enforced at all times
- Operations
 - Specify database retrievals and updates
 - May include basic model operations (generic insert, delete, and update) and user-defined operations (compute_student_gpa, update_inventory)

Categories of Data Models

- Conceptual data models (high-level)
 - Provide concepts that are close to the way many users perceive data
- Physical data models (low-level)
 - Provide concepts that describe details of how data is stored in the computer
- Representation data models (implementation)
 - Provide concepts that fall between the above two – may be easily understood by end users but not too far from the way data is organized
 - Include widely used relational data model and object data model
- Self-Describing Data Models
 - NoSQL database

Database Schemas

- The description of a database

STUDENT

Name	Student_number	Class	Major
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COURSE

Course_name	Course_number	Credit_hours	Department
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PREREQUISITE

Course_number	Prerequisite_number
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SECTION

Section_identifier	Course_number	Semester	Year	Instructor
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GRADE_REPORT

Student_number	Section_identifier	Grade
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Database State (Instance)

- The data in the database at a particular moment (snapshot)

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CS3380	MATH2410
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Schemas VS. States

- Database schema
 - Structure, data types, and constraints
 - Changes very infrequently
- Database state
 - Changes every time the database is updated

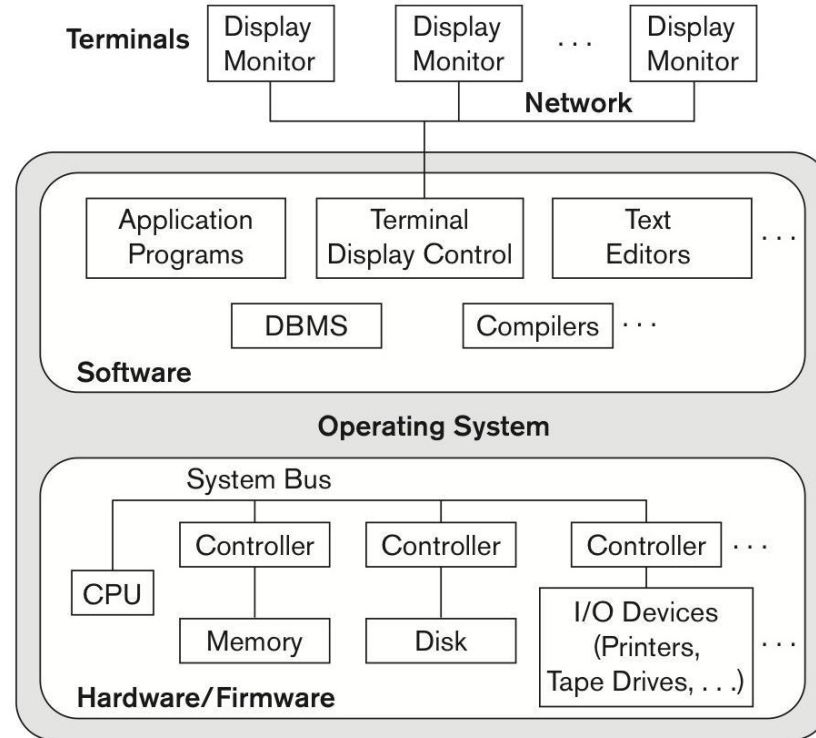
DBMS Languages

- DDL (Data Definition Language)
 - Define databases, tables, fields, and index
 - In SQL: **CREATE**, **DROP**, **ALTER**
- DML (Data Manipulation Language)
 - Specify database retrievals and updates
 - In SQL: **SELECT**, **INSERT**, **UPDATE**, **DELETE**

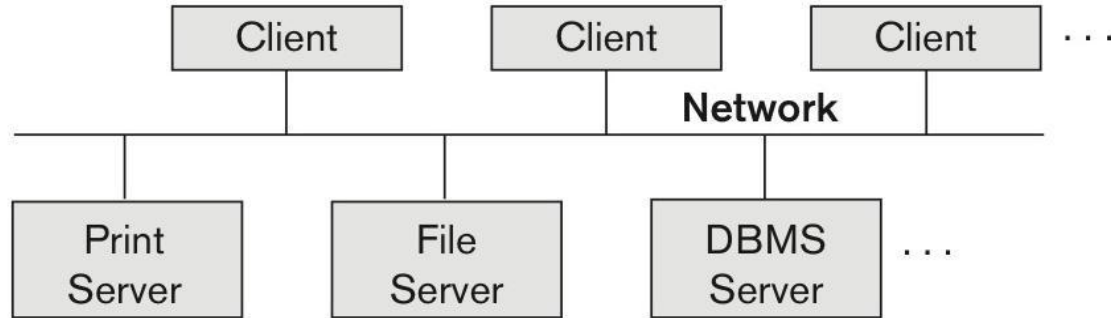
DBMS Architecture

- Centralized DBMS architecture
- Client/Server architecture

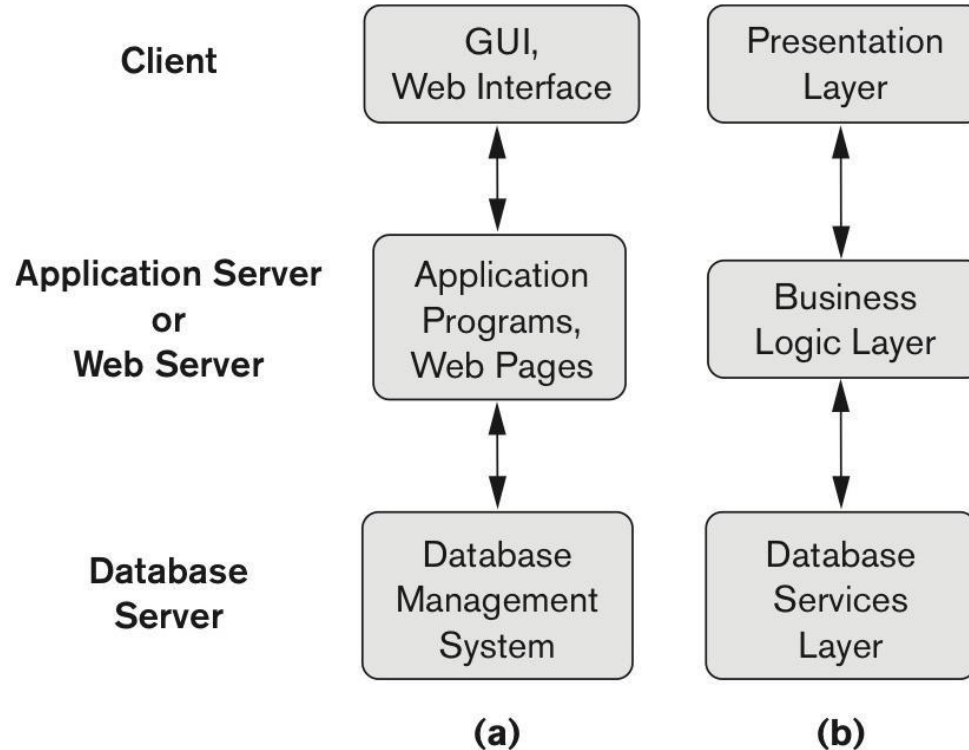
Centralized DBMS Architecture



Logical Two-tier Client/Server Architecture



Logical Three-tier Client/Server Architecture



Discussion

- If you were designing a system to book and sell airline tickets, which DBMS Architecture would you choose?
- What happens if a different architecture is used?

Database Users

- Actors on the scene
 - Those who use and control the database content, and those who design, develop and maintain database applications
- Workers behind the scene
 - Those who design and develop the DBMS software and related tools, and the computer systems operators but are not actively interested in the database contents

Actors On The Scene

- Database administrators
 - Responsible for authorizing access to the database, coordinating and monitoring its use, acquiring software and hardware resources, controlling its use, and monitoring the efficiency of operations
- Database designers
 - Responsible for defining the content, the structure, the constraints, and functions or transactions against the database. They must communicate with the end-users and understand their needs.
- End users
 - They use the data for queries and reports, and some of them update the database content.
- System analysts and application programmers

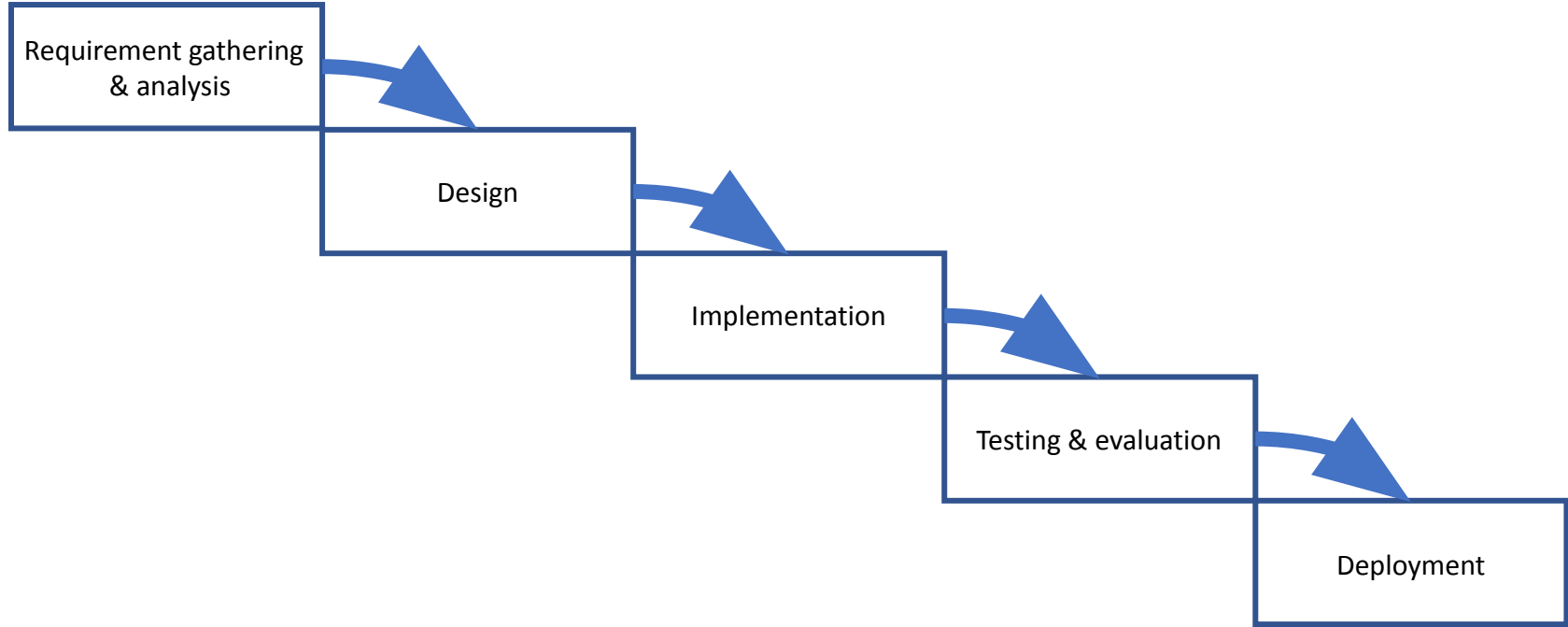
Workers Behind The Scene

- System Designers and Implementors
 - Design and implement DBMS packages in the form of modules and interfaces and test and debug them. The DBMS must interface with applications, language compilers, operating system components, etc.
- Tool Developers
 - Design and implement software systems called tools for modeling and designing databases, performance monitoring, prototyping, test data generation, user interface creation, simulation, etc. that facilitate the building of applications and allow using databases effectively.
- Operators and Maintenance Personnel
 - They manage the actual running and maintenance of the database system hardware and software environment.

Database Development

- Requirements gathering and analysis
- Design
- Implementation
- Testing and evaluation
- Deployment

Waterfall Process Model



Requirements gathering and analysis

- Understand the overall purpose and scope of the software system
- Assign database specialists to the team
- Identify high-level requirement
- Refine into detailed requirements

Design

- Requirements → conceptual design/schema
 - Identify things/entities (nouns)
 - Identify relationships among things (verbs)
 - Identify attributes (nouns, adjectives)
- Conceptual schema → logical design/schema
 - Map conceptual model to implementation model
- Logical design schema → physical design/schema
 - Choose suitable DBMS
 - Write constructs in chosen DBMS (DDL)

Implementation

- Install and configure DBMS
- Create development and test database (DDL)
- Create tables and other constructs (DDL)
- Load data into databases (DML)
- Set up users and security

Testing and evaluation

- Test the performance
- Test security aspects
- Test data integrity
- Perform load and stress testing
- Fine-tune parameters

Deployment

- Create production database
- Perform data conversion and loading